

From Cross-Sections to Panels

Session 5

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Reading

Deaton is in your reading/ folder on the hub, or PDF.

- Deaton, ch. 2 §2.7 and ch. 6
- Mundlak (1961), "Empirical production function free of management bias" *J. of Farm Economics* 43(1):44-56.
- Imbens & Wooldridge (2009), "Linear Panel Data Models," cemmap lectures 3–4 (slides I, slides II, notes). Notes sections 1, 3, and 6: fixed effects and first differences, exogeneity, and pseudo-panels.

Reading: cohorts and inference

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton & Paxson (1994), "Intertemporal Choice and Inequality," *J. of Political Economy* 102(3):437–467 (paper).
Section II: following birth cohorts and their consumption dispersion.
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

Longitudinal Panels

Introduction to Panels

A panel follows the *same* households across survey waves.

- i identifies a household; t identifies a wave.
- y_{it} is an outcome; z_{it} is a vector of observed covariates.
- Changes within a household let us hold its persistent characteristics fixed, even when we cannot measure them.
- Repeated cross-sections follow a population; a panel also follows its members. Matching household identifiers is what makes the difference.

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Why two sets of effects

Across farms, an input's coefficient carries the farmer.
Mundlak (1961).

- *The manager you cannot see.* Correlated with every input the farm chooses. Follow the same farm and it cancels.
- *The year everyone shared.* 2008 doubled cereal prices for everyone. Charge that to a household and you have measured the market.

Later, on 506 Ghanaian farms:

Log input	Across farms	Within farms
Crop area	0.660	0.363
Labor	0.297	0.091

Two-way Fixed Effects Regressions

Consider a common-slope model:

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}.$$

- α_i : persistent household differences, such as ability or land quality. These may be correlated with z_{it} .
- δ_t : shocks shared by every household in a wave.
- β : identified by changes in z_{it} within households, after removing common time movements.

A time-invariant covariate is absorbed by α_i ; a covariate that varies only by wave is absorbed by δ_t .

TWFE generalizes a double difference

Two groups, A and B ; two dates, before (0) and after (1). Apply the TWFE model to their mean outcomes, with treatment only in $(A, 1)$. Its treatment coefficient is

$$\hat{\beta} = (\bar{y}_{A1} - \bar{y}_{A0}) - (\bar{y}_{B1} - \bar{y}_{B0}).$$

- Difference over time: each group's fixed effect cancels.
- Difference across groups: the common time change cancels.
- TWFE extends this to more units, dates and covariates.

Parallel *untreated* trends is the two-group, two-date form of strict exogeneity below — not something the double difference owns and the regression lacks.

The within transformation

The same two subtractions, for any number of households and waves, with $\bar{q}_{i\cdot}$ a household's mean, $\bar{q}_{\cdot t}$ a wave's mean, and $\bar{q}_{\cdot\cdot}$ the grand mean:

$$\tilde{q}_{it} = q_{it} - \bar{q}_{i\cdot} - \bar{q}_{\cdot t} + \bar{q}_{\cdot\cdot}$$

- The household mean removes α_i , as differencing over time did.
- The wave mean removes δ_t , as differencing across groups did.
- Both carry off the grand mean, so it returns once.

Apply it to y and every covariate; $\hat{\beta}$ is then the OLS slope of $\tilde{y}$ on $\tilde{z}$ — so long as every household appears in every wave.

You have already used one with different slopes

Session 4's demand system is this model with the common slope relaxed:

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}, \quad \log x_{it}^j = a_t^j + \beta_j w_{it} + \gamma_j' d_{it} + \varepsilon_{it}^j.$$

- δ_t becomes a_t^j : the wave effect differs by good.
- The household term is neither fixed nor equally loaded: w_{it} moves with t , and each good answers it through its own β_j .
- Set $\beta_j = \beta$, $w_{it} = w_i$ and it collapses back to an additive household effect.

The index j buys this: differing slopes identify w from the *pattern* across goods, with no prices and no total expenditure. And note which term is the nuisance — here it is the estimand.

Where the causal content comes from

The estimator removes α_i and δ_t . That is all it does, and it does it whatever z is.

- Whether β is causal turns on how z came to vary, never on which subtraction you performed.
- Randomize z and strict exogeneity holds by construction: a property of z , with the estimator unchanged.
- A double difference has no warrant a TWFE regression lacks.
- With an observational z you owe an argument, either way.

The design supplies the exogeneity; the estimator supplies the subtraction. Do not read the second as the first.

What makes the within comparison valid?

A sufficient *strict exogeneity* condition is

$$\mathbb{E}[\epsilon_{it} \mid z_{i1}, \dots, z_{iT}, \alpha_i] = 0 \quad \text{for every } t.$$

- Covariates at *all* dates must be unrelated to the idiosyncratic shock at t . Feedback into future covariates can violate this.
- After removing both effects, the covariates must still have full column rank.
- Large-sample arguments require many independent sampling units and suitable moment conditions.

Allow for serial correlation within households and dependence within sampling clusters when computing standard errors.

What fixed effects can remove

- A time-constant omitted household characteristic: absorbed by α_i , if it enters additively.
- A common additive time shock: absorbed by δ_t .
- A persistent additive reporting error: also absorbed, if its contribution to the equation stays constant.

What remains a problem:

- household-specific shocks correlated with changing covariates;
- time-varying effects of an omitted characteristic;
- time-varying measurement error in the regressors.

Within variation can contain proportionally *more* measurement error.

Attrition

Attrition is loss of intended follow-up observations. It is not every decline in the number of household IDs shared by two tables.

- Planned rotation removes households by design.
- Refreshment adds newly sampled households.
- Splits change the household being followed.
- Missing modules can remove observations from a regression even when the household was interviewed.

Read the sampling and tracking rules before calling an overlap loss an attrition rate.

Attrition and consistency

Let $R_{it} = 1$ when the observation is available, and let $R_i = (R_{i1}, \dots, R_{iT})$. A sufficient condition for the common-slope model is

$$\mathbb{E}[\epsilon_{it} \mid z_{i1}, \dots, z_{iT}, \alpha_i, R_i] = 0.$$

- Selection may depend on α_i and on the covariates.
- Selection driven by idiosyncratic outcome shocks can violate this condition, even if the shock precedes departure.
- We still need identifying variation in the observed sample.

Keeping only households observed in every wave does not repair selection on shocks.

A genuine panel: Uganda

The Uganda National Panel Survey draws its initial households from the 2005–06 household survey. The library carries eight time labels through 2019–20.

- Continuing households identify within-household changes.
- Split-offs and refreshment households also enter the data.
- Sample refreshment began in 2013–14; a decline in overlap across that boundary includes a design change.

We first examine 2009–10, 2010–11, and 2011–12, before that refresh. UBOS sampling and tracking description.

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Removing Fixed Effects: a balanced panel

A *balanced* panel observes every household in the same T waves. An *unbalanced* panel observes different subsets of waves.

Balanced and equally weighted is the case the within transformation was written for. Applied to y and every covariate it leaves

$$\tilde{y}_{it} = \beta' \tilde{z}_{it} + \tilde{\epsilon}_{it}.$$

Both sets of intercepts are gone, so regress without an intercept.

The attrition matrix says Uganda is not balanced, and neither is GLSS. So this is not the estimator either of them gets.

Removing Fixed Effects: an unbalanced panel

Household demeaning still removes α_i , using each household's observed waves. Subtracting the two sets of means *once* generally does not remove both effects.

Let D contain household and time indicators on the observed rows:

$$M_D = I - D(D'D)^+D', \quad \hat{\beta} = (Z'M_DZ)^{-1}Z'M_Dy.$$

This is Frisch–Waugh–Lovell:

1. Remove the fitted values from projecting y and each column of Z on the *same* D .
2. Regress the residual outcome on the residual covariates.

A GLSS panel

GLSS1 (1987–88) and GLSS2 (1988–89) contain a two-wave panel.

- Half of GLSS1's sampling workloads were retained for GLSS2; half were replaced.
- PANELC.DAT provides person-level links. The library applies these, keeping split-off households distinct.
- Later GLSS rounds have no registered household linkage.
- Loss of overlap includes planned rotation, not just missed follow-up interviews.

GSS, *GLSS3 Report*, Appendix 1, p. 110.

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Application: Farm Management in Ghana

Management and measured productivity

Farms with more inputs often produce more. They may also be managed differently.

- A farmer who chooses better planting dates may also cultivate more land or hire more labor.
- A cross-sectional regression can attribute that persistent advantage to the measured inputs.
- Repeated observations let us compare a farm with itself, holding fixed an additive management effect.

Mundlak (1961) motivates this use of farm effects. We apply the idea to household crop values in the two-wave GLSS panel.

Cobb–Douglas: technology and management

For one output and positive inputs, suppose

$$Q_{it} = A_{it} H_{it}^{\beta_H} L_{it}^{\beta_L} K_{it}^{\beta_K}.$$

- H : land; L : labor; K : another productive input.
- A : productivity, including management and unmeasured conditions.
- Each β is a physical-output elasticity in this model.
- Scaling all inputs scales output by the factor raised to $\beta_H + \beta_L + \beta_K$.

The form describes a technology. It doesn't say inputs are chosen independently of productivity.

From output to a linear revenue equation

No harvest quantity is recorded, so the left side is revenue.

With $R_{it} = p_{it}Q_{it}$,

$$\log R_{it} = \beta_H \log H_{it} + \beta_L \log L_{it} + \underbrace{\log p_{it} + \log A_{it} + \beta_K \log K_{it}}_{u_{it}}.$$

- Value out, quantities in: the output price has nowhere to go but u .
- K is observed as *spend*, not quantity, and for 256 of 506 farms.
- Correlation between u and the measured inputs biases OLS, and a sum of crop values needn't have this single-output structure anyway.

So this describes *partial crop value*, not physical output.

What the Ghana tables observe

- Output: values sold, kept for seed, given away, and lost; own consumption is absent.
- Land: harvested area summed across crops.
- Labor: family hours, hired expenditure, exchange days.
- Inputs: recorded spending, with marketing costs separate.

Harvest quantities are unavailable. Valuations mix sale and replacement prices. We won't infer missing output from land.

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Cross-sectional estimation

At a single date, regress log crop value on log area and log labor:

$$y_i = c + \beta_H h_i + \beta_L \ell_i + u_i.$$

- Management, omitted inputs, and prices enter u_i .
- OLS needs this residual uncorrelated with the regressors.
- Productive farmers may choose more inputs: coefficients can combine input effects with selection across farms.

Compare each wave with changes among the *same* households.

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Two waves: remove persistent omitted factors

Suppose the composite omitted term is $u_{it} = \alpha_i + \delta_t + \epsilon_{it}$. Then

$$\Delta y_i = (\delta_2 - \delta_1) + \beta_H \Delta h_i + \beta_L \Delta \ell_i + \Delta \epsilon_i.$$

- Persistent management, land quality, and price differences cancel *if their contributions are additive and constant*.
- The intercept captures the common time change.
- Shocks that change input choices, changing prices, and measurement error remain potential sources of bias.

We need $\mathbb{E}[\Delta \epsilon_i \mid z_{i1}, z_{i2}, \alpha_i] = 0$ in the estimation sample, plus identifying variation.

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What changes when we follow farms?

Same 506 farms; nominal partial crop value; unweighted estimates:

Log input	Pooled + wave	First differences
Crop area	0.660	0.363
Labor	0.297	0.091

- Both slopes fall, and land falls by nearly half.
- Partial output and unstable area remain problems.

This doesn't isolate a causal management effect.

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Lifecycle consumption

Why study consumption over a lifetime?

Income and needs change over a lifetime. Consumption needn't move one-for-one with current income.

- Saving transfers resources from today to later ages.
- Borrowing transfers future resources toward today, when credit is available.
- Uncertain earnings, health, and longevity affect those choices.

The life-cycle model asks how households allocate resources across ages. It motivates an age profile of consumption; it doesn't determine that profile without assumptions about preferences and constraints.

The life-cycle budget

At age a , choose consumption c_a and assets to carry forward:

$$b_{a+1} = (1 + r)b_a + y_a - c_a.$$

Here b_a is assets, y_a income, and r the interest rate.

- Choices trade current against expected future utility, given initial assets and a requirement to repay debts by the end of life.
- Credit limits can prevent the desired transfer of resources.
- Uninsured shocks can make households' consumption paths diverge.

We need data across ages to study these implications.

An application: inequality as cohorts age

Deaton and Paxson (1994) follow consumption inequality among people born around the same time. Our question is narrower:

Does dispersion in food purchases per household member rise as a cohort ages?

- The mean describes the cohort's average log purchases.
- The variance describes differences *among* its households.
- We compare the same birth cohorts across survey waves.

Food purchases omit own-produced food and non-food consumption. This exercise doesn't measure total consumption inequality.

Panels from Cross-Sections

Following groups instead of households

Most GLSS rounds interview a new sample of households. We can't compute an individual household's change between those rounds.

We can follow a *birth cohort*: for example, households whose heads were born in 1960–64.

- Find representatives of that cohort in each survey.
- Summarize their circumstances using the survey weights.
- Follow those summaries as the cohort ages.

The group definition stays fixed; the sampled households change.

A pseudo-panel

For cohort c in wave t , estimate the population mean with

$$\bar{y}_{ct} = \frac{\sum_{i \in (c,t)} w_{it} y_{it}}{\sum_{i \in (c,t)} w_{it}}.$$

- One row now describes a cohort and a date.
- Repeated rows let us study changes in cohort means or dispersion.
- Small cells give noisy summaries, even when the whole survey is large.

This is a panel of *estimated group characteristics*, not a household panel. It can't reveal which particular households moved up or down.

From cells to a regression

A working model for cohort means is

$$\bar{y}_{ct} = \alpha_c + \delta_t + \beta' \bar{z}_{ct} + u_{ct}.$$

Age terms and household composition can enter $\bar{z}_{ct}$.

- α_c captures persistent differences between cohorts.
- δ_t captures changes common to cohorts at a survey date.
- Both outcome and regressor means contain sampling error.

Larger cells help, but don't establish that the regression error is unrelated to the regressors. We'll first inspect the cohort profiles.

Identification: age, cohort, and time

For a birth year b , age a , and calendar year t ,

$$a = t - b.$$

The data cannot assign a linear trend uniquely to age, cohort, or time.

- Following a cohort changes both its age and the calendar year.
- Comparing ages in one survey also compares birth cohorts.
- Separating all three requires an explicit restriction.

Cohort paths aren't pure age effects.

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Household panels and welfare

Exercise 1: ten-year cohorts

Rebuild the Ghana pseudo-panel with ten-year cohorts. How much do the lifecycle profiles change? Which conclusion is robust?

Exercise 2: means of logs, logs of means

Compare the weighted mean of log purchases with the log of weighted mean purchases in each cohort. Why are they different?

Exercise 3: attrition in Uganda

What fraction of the 2009–10 households appear in 2011–12?
Are the leavers different in 2009–10 consumption from the stayers? What does that do to your fixed-effects estimate?

Exercise 4: Mundlak (1978) on the farms

Mundlak's later formulation augments the *levels* regression with farm means of the regressors. Add them, keep the wave indicator, and show the slopes reproduce the first differences — an algebra check on the implementation, good to machine precision.

Then test the mean terms jointly with the farm-cluster covariance. What does rejection establish, and what does it not?

Exercise 5: FWL on an unbalanced panel

Redo the within estimator by Frisch–Waugh–Lovell on an unbalanced panel of your own making: eight farms, four dates, a few rows deleted. Use `DataMat.resid` to partial the farm and wave indicators out of the outcome and the regressors *jointly*, then regress the residuals.

Check it against full indicator OLS, and check that one pass of subtracting farm means and wave means does *not* agree. Why does balance matter here when it didn't in the balanced case? (`metrics_miscellany.estimators.fwl_regression` partials recursively.)

Exercise 6: put the purchased inputs back

Purchased inputs sit in u . Add $\log$ production-input spend as a third regressor. Only 256 of the 506 farms report positive spend in both waves, so re-estimate the two-input model on those same 256 as well — three sets of slopes, not two.

How much of the movement in β_H and β_L is the omitted input, and how much is the change of sample? And what must be true of input prices for the new coefficient to be an elasticity rather than a description?